NOT MEASUREMENT SENSITIVE

MIL-DTL-25576F
16 March 2023

SUPERSEDING
MIL-DTL-25576E
14 April 2006

DETAIL SPECIFICATION

PROPELLANT, ROCKET GRADE KEROSENE

This specification is approved for use by
all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two grades of rocket propellant kerosene for use in rocket engines.

1.2 Classification. Rocket propellant kerosene will be of the following designated grades:

RP-1	-	Normal production, total sulfur content of 30 mg/kg (max) and 2 percent by volume (max) of olefins, suitable for most uses.
RP-2	-	Processed RP-1, total sulfur content of 100 µg/kg (max) and 1 percent by volume (max) of olefins.

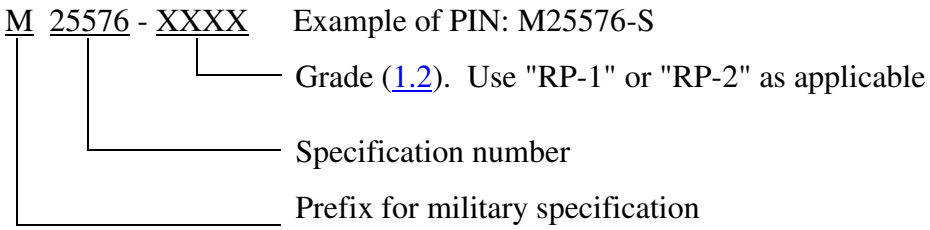
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1.3 Part or identifying number (PIN). The PINs to be used for rocket grade propellant kerosene acquired to this specification are created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-290 - Packaging and Marking of Petroleum and Related Products

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL (ASTM)

ASTM D56	-	Standard Test Method for Flash Point by Tag Closed Cup Tester
ASTM D86	-	Standard Test Method for Distillation of Petroleum Products and Liquid Fuels at Atmospheric Pressure
ASTM D93	-	Standard Test Methods for Flash Point by Pensky-Martens Closed Cup Tester
ASTM D130	-	Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test

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ASTM D240	-	Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter
ASTM D381	-	Standard Test Method for Gum Content in Fuels by Jet Evaporation
ASTM D445	-	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D1298	-	Standard Method for Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer method
ASTM D1319	-	Standard Test Method for Hydrocarbon Types in Liquid Petroleum Products by Fluorescent Indicator Adsorption
ASTM D2276	-	Standard Test Method for Particulate Contamination in Aviation Fuel by Line Sampling
ASTM D2386	-	Standard Test Method for Freezing Point of Aviation Fuels
ASTM D2622	-	Standard Test Method for Determination of Sulfur in Petroleum Products by Wavelength Dispersive X-Ray Fluorescence Spectrometry
ASTM D3227	-	Standard Test Method for (Thiol Mercaptan) Sulfur in Gasoline, Kerosine, Aviation Turbine, and Distillate Fuels (Potentiometric Method)
ASTM D3241	-	Standard Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels
ASTM D3338/D3338M	-	Standard Test Method for Estimation of Net Heat of Combustion of Aviation Fuels
ASTM D3343	-	Standard Test Method for Estimation of Hydrogen Content of Aviation Fuels
ASTM D3701	-	Standard Test Method for Hydrogen Content of Aviation Turbine Fuels by Low Resolution Nuclear Magnetic Resonance Spectrometry
ASTM D3828	-	Standard Test Methods for Flash Point by Small Scale Closed Cup Tester
ASTM D4045	-	Standard Test Method for Sulfur in Petroleum Products by Hydrogenolysis and Rateometric Colorimetry
ASTM D4052	-	Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter
ASTM D4057	-	Standard Practice for Manual Sampling of Petroleum and Petroleum Products

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ASTM D4177	-	Standard Practice for Automatic Sampling of Petroleum and Petroleum Products
ASTM D4306	-	Standard Practice for Aviation Fuel Sample Containers for Tests Affected by Trace Contamination
ASTM D4809	-	Standard Test Method for Heat of Combustion of Liquid Hydrocarbon Fuels by Bomb Calorimeter (Precision Method)
ASTM D4952	-	Standard Test Method for Qualitative Analysis for Active Sulfur Species in Fuels and Solvents (Doctor Test)
ASTM D 5452	-	Standard Test Method for Particulate Contamination in Aviation Fuels by Laboratory Filtration
ASTM D 5453	-	Standard Test Method for Determination of Total Sulfur in Light Hydrocarbons, Spark-Ignition Engine Fuel, Diesel Engine Fuel, and Engine Oils by Ultraviolet Fluorescence
ASTM D5972	-	Standard Test Method for Freezing Point of Aviation Fuels (Automatic Phase Transition Method)
ASTM D6379	-	Standard Test Method for Determination of Aromatic Hydrocarbon Types in Aviation Fuels and Petroleum Distillates—High Performance Liquid Chromatography Method with Refractive Index Detection
ASTM D6986	-	Standard Test Method for Free Water, Particulate and Other Contamination in Aviation Fuels (Visual Inspection Procedures)
ASTM D7039	-	Standard Test Method for Sulfur in Gasoline, Diesel Fuel, Jet Fuel, Kerosine, Biodiesel, Biodiesel Blends, and Gasoline-Ethanol Blends by Monochromatic Wavelength Dispersive X-ray Fluorescence Spectrometry
ASTM D7042	-	Standard Test Method for Dynamic Viscosity and Density of Liquids by Stabinger Viscometer (and the Calculation of Kinematic Viscosity)
ASTM D7094	-	Standard Test Method for Flash Point by Modified Continuously Closed Cup (MCCCFP) Tester
ASTM D7153	-	Standard Test Method for Freezing Point of Aviation Fuels (Automatic Laser Method
ASTM D7154	-	Standard Test Method for Freezing Point of Aviation Fuels (Automatic Fiber Optical Method)
ASTM D7171	-	Standard Test Method for Hydrogen Content of Middle Distillate Petroleum Products by Low-Resolution Pulsed Nuclear Magnetic Resonance Spectroscopy

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ASTM D7236	-	Standard Test Method for Flash Point by Small Scale Close Cup Tester (Ramp Method)
ASTM D7345	-	Standard Test Method for Distillation of Petroleum Products and Liquid Fuels at Atmospheric Pressure (Micro Distillation Method)
ASTM D7619	-	Standard Test Method for Sizing and Counting Particles in Light and Middle Distillate Fuels, by Automatic Particle Counter
ASTM D7777	-	Standard Test Method for Density, Relative Density, or API Gravity of Liquid Petroleum by Portable Digital Density Meter
ASTM D7945	-	Standard Test Method for Determination of Dynamic Viscosity and Derived Kinematic Viscosity of Liquids by Constant Pressure Viscometer
ASTM D8049	-	Standard Test Method for Determining Concentration, Count, and Size Distribution of Solid Particles and Water in Light and Middle Distillate Fuels by Direct Imaging Analyzer
ASTM D8072	-	Standard Classification for Reporting Solids and Insoluble Water Contamination of Hydrocarbon-Based Petroleum Products When Analyzed by Imaging Instrumentation
ASTM D8305	-	Standard Test Method for The Determination of Total Aromatic Hydrocarbons and Total Polynuclear Aromatic Hydrocarbons in Aviation Turbine Fuels and other Kerosene Range Fuels by Supercritical Fluid Chromatography
ASTM E29	-	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

(Copies of these documents are available from https://www.astm.org/.)

ENERGY INSTITUTE (EI)

EI 1581	-	Specifications and laboratory qualification procedures for aviation fuel filter/water separators
IP 540	-	Determination of the existent gum content of aviation fuel – Jet evaporation method
IP 565	-	Determination of the level of cleanliness of aviation turbine fuel – Portable automatic particle counter method
IP 577	-	Determination of the level of cleanliness of aviation turbine fuel – Automatic particle counter method using light extinction

(Copies of these documents are available from https://publishing.energyinst.org.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4406 - Hydraulic fluid power — Fluids — Method for coding the level of contamination by solid particles

ISO 11171 - Hydraulic Fluid Power – Calibration of Automatic Particle Counters for Liquids

(Copies of these documents are available from https://www.iso.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The propellant shall consist completely of hydrocarbon compounds except as otherwise specified herein.

3.2 Chemical and physical properties. The chemical and physical properties of the propellant shall conform to those listed in Table I when tested in accordance with applicable test methods.

TABLE I. Chemical and physical properties and test methods.

Property	Limits		ASTM or IP Test Method
	RP-1	RP-2	
Workmanship	Pass	Pass	Section 3.6
Distillation temperature, °F (°C)			D86 [1], D7345 [2]
Initial Boiling Point, °F (°C)	Report Only	Report Only	
10 percent recovered, °F (°C)	365 to 410 (185 to 210)	365 to 410 (185 to 210)	
50 percent recovered, °F (°C)	Report Only	Report Only	
90 percent recovered, °F (°C)	Report Only	Report Only	
Final boiling point, °F (°C)	max 525 (274)	max 525 (274)	
Residue, vol. percent	max 1.5	max 1.5	
Loss, vol. percent	max 1.5	max 1.5	
Specific Gravity, 60/60°F	0.799 – 0.815	0.799 – 0.815	D1298 [1], D4052, D7777
Existent Gum, mg/100mL	max 1	max 1	D381 [1,3] , IP 540 [1]

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TABLE I. Chemical and physical properties and test methods - Continued.

Property	Limits		ASTM or IP Test Method
	RP-1	RP-2	
Sulfur, total, mass percent (mg/kg (ppm by mass))	max 0.003 (30)	max 0.00001 (0.1)	D5453 [4], D2622 [4], D4045 [4], D7039 [4]
Sulfur, mercaptan, mass percent (mg/kg (ppm by mass))	max 0.0003 (3)	[5]	D3227 [1], D4952 [6]
Freezing Point, °F (°C)	max -60 (-51)	max -60 (-51)	D2386 [1], D5972, D7153, D7154
Net heat of combustion, BTU/lb (MJ/kg)	min 18500 (43.0)	min 18500 (43.0)	D240, D3338 [7], D4809 [1]
Viscosity at -30°F (-34°C), cSt (mm^2/s)	max 16.5	max 16.5	D445 [1], D7042 [8], D7945
Aromatics, vol. percent	max 5.0	max 5.0	D1319 [1], D6379, D8305 [9]
Olefins, vol. percent	max 2	max 1	D1319
Hydrogen content, mass percent	min 13.8	min 13.8	D3343, D3701, D7171 [1]
Copper Strip Corrosion (2 hr at 100°C), rating	max 1	max 1	D130 [10]
Microseparometer Rating	min 90	min 90	D3948
Flash Point, °F (°C) [11]	min 140 (60)	min 140 (60)	D93 [1], D56, D3828, D7236, D7094 [12]
Thermal Stability (2.5 hr at 617°F (325°C))			D3241
Change in pressure drop, mm Hg	max 10	max 10	
Tube rating: One of the following shall be measured			
(1) Annex A1 VTR	<3 [13]	<3 [13]	
(2) Annex A2, ITR [14], nm	max 50	max 50	
(3) Annex A3, ETR [1], nm	max 50	max 50	
Particulate Matter: One of the following shall be measured [15]			
Gravimetric, mg/L	max 1.0	max 1.0	D5452 [1], D2276
Particle counting [16], ISO Code	max 19/17/14/13	max 19/17/14/13	IP 565, IP 577, D7619 [1], D8049 [17]

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TABLE I. Chemical and physical properties and test methods - Continued.

Notes:	
1.	Referee Test Method.
2.	ASTM D7345 results shall not be corrected to bias-free distillation results by the application of the correction factors in the D7345 Precision and Bias section.
3.	The preferred vaporizing medium for the propellant is steam; however, the existent gum test IP 540 may be performed using air as the vaporizing medium. If air is used instead of steam, it shall be recorded. Test Method ASTM D381, using steam jet operating conditions, shall be the referee test method.
4.	ASTM D2622, D5453 (referee for RP-1), or D7039 may be used for RP-1 grade. ASTM D4045 shall be used for RP-2 grade.
5.	Not required.
6.	If the Doctor Test results in a failure ('positive' result), then mercaptan sulfur content shall be determined by the referee test method ASTM D3227.
7.	ASTM D3338 and ASTM D3338/D3338M are the same test method.
8.	ASTM D7042 results shall be corrected to bias-free viscosity results by the application of the correction factor in the D7042 Precision and Bias section.
9.	Results from ASTM D8305 shall be bias-corrected using the bias-correction equation for total aromatics in Section 13 (Precision and Bias) of ASTM D8305.
10.	Follow the procedures specific to kerosene.
11.	Relative to ASTM D93: results obtained by D56 can be up to 1.5°C lower, results obtained by D3828 can be up to 2°C lower, results obtained by D7094 are typically the same as D93, and results obtained by D7236 can be up to 1.0 °C lower.
12.	Use of ASTM D7094 is limited to non-procurement or operational quality surveillance related analysis only.
13.	Peacock or Abnormal color deposits result in a failure.
14.	If the ITR reports "N/A" for the tube's volume measurement, the result constitutes a failure and the result shall be reported as ">50 nm."
15.	Particulate matter shall be determined by either gravimetric or particle counting. Should particle counting result in failure, the gravimetric method shall be the referee.

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TABLE I. Chemical and physical properties and test methods - Continued.

<table>
<tr><td colspan="2">Notes:</td></tr>
<tr><td>16.</td><td>Particle counting, where equipment and laboratory capability exist, is the alternate to the gravimetric method. The limits provided below are for particulate matter for procurement requirements only. Should particle counting result in failure, the gravimetric method shall be the referee.
<table>
<tr><th>Particle counting, individual channel counts & ISO Code</th><th>Channel Counts</th><th>ISO Code 16a</th><th>Test Method IP 565, IP 577, or D7619</th></tr>
<tr><td>≥ 4 µm (c)16b</td><td>16c</td><td>19</td><td></td></tr>
<tr><td>≥ 6 µm (c)16b</td><td>16c</td><td>17</td><td></td></tr>
<tr><td>≥ 14 µm (c)16b</td><td>16c</td><td>14</td><td></td></tr>
<tr><td>≥ 30 µm (c)16b</td><td>16c</td><td>13</td><td></td></tr>
</table>
16a. Per Table 1 "Allocation of scale numbers" of ISO 4406.
16b. (c) indicates that the equipment has been calibrated in accordance with ISO 11171.
16c. Report.</td></tr>
<tr><td>17.</td><td>If ASTM D8049 is used, the maximum reporting code would be ASTM D8049 19/17/14-10. The reporting format for the code shall be IAW ASTM D8072 and reported as DXXXX X/Y/Z-W where DXXXX details the ASTM standard used to perform the analysis; X, Y, and Z are identifying numbers from Table 1 of ASTM D8072 for particle counts in the ≥4 µm, ≥6 µm, and ≥14 µm ranges respectively; and W is the identifying number for the water concentration in ppm by volume.</td></tr>
</table>

3.3 Additives. The additives listed herein may be used singly or in combination, in amounts not to exceed those specified. No substance of known dangerous toxicity under usual conditions of handling and use shall be added except as specified herein. The type and amount of each additive used shall be reported and shall be specified in the contract or order.

3.3.1 Antioxidants. The following active inhibitors may be added separately or in combination to the propellant in total concentration not in excess of 8.4 pounds of inhibitor (not including weight of solvent) per 1,000 barrels (6.3.1) of propellant (9.1g /100 US gal) in order to prevent the formation of gum.

a. 2,6-ditertiary butyl 4-methyl phenol.

b. 2,4-dimethyl-6 tertiary-butyl phenol.

c. 2,6-ditertiary butyl phenol.

3.3.2 Metal deactivator. A metal deactivator (MDA), N, N'-disalicylidene-1, 2-propanediamine, may be added in an amount not to exceed 2 pounds of active ingredient per 1,000 barrels of propellant (2.2 g/100 US gal). Use of MDA should be limited to only when metal contamination has been proven and that after the use of MDA, propellant should be clay treated prior to use to remove the MDA and metal.

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3.4 Limiting values. For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off to the nearest unit in the last right-hand digit used in expressing the specified limit. This rounding off shall be done in accordance with the Rounding Method of ASTM E29.

3.5 Filter. A filter water separator conforming to the requirements of the latest revision of EI 1581, with at minimum Category C elements, shall be installed on the fill line upstream of the header used for filling the delivery containers.

3.6 Workmanship. At the time of Government acceptance, the finished propellant shall be visually free from undissolved water, sediment, or suspended matter, and shall be clear and bright (no haze or cloudiness). Refer to ASTM D6986. In case of a dispute, the propellant shall be clear and bright at 70°F.

4. VERIFICATION

4.1 Classification of inspections. The inspection and testing of the propellant requirements specified herein are classified as conformance tests (4.2).

4.2 Conformance inspection conditions. Tests for acceptance of individual lots shall consist of tests for all requirements specified in section 3. Conformance inspection shall include the test requirement herein.

4.2.1 Inspection lot. For acceptance purposes, individual lots shall be examined as specified herein and subject to tests for all requirements cited in section 3.

4.3 Inspection.

4.3.1 Inspection conditions. Propellant supplied shall comply with the specified limiting values in Table I using the cited test methods. The specified limiting values must not be changed. This precludes any allowance for test method precision and adding or subtracting digits. For the purposes of determining conformance with the specified limiting values, an observed value or a calculated value shall be rounded off “to the nearest unit” in the last right-hand place of digits used in expressing the specified limiting value, in accordance with the Rounding Method of ASTM E29.

4.4 Sampling plans.

4.4.1 Sampling. Each bulk or packaged lot of propellant shall be sampled for verification of product conformance in accordance with ASTM D4057 or ASTM D4177, except where individual test procedures contain specific sampling instructions.

4.4.1.1 A number of propellant properties are very sensitive to trace contamination that can originate from sample containers. For recommended sample containers, refer to ASTM D4306.

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4.4.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected from each lot and shall be subjected to the examination of filled containers as specified in 4.5.1.3.

4.5 Methods of inspection.

4.5.1 Examination of product.

4.5.1.1 Visual inspection. Samples selected in accordance with 4.4.1 shall be visually examined for compliance with 3.6.

4.5.1.2 Examination of empty containers. Before filled, each unit container shall be visually inspected for cleanliness and suitability in accordance with ASTM D4057.

4.5.1.3 Examination of filled containers. Samples taken as specified in 4.4.2 shall be examined for conformance to MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and markings. Any container with one or more defects under the required fill shall be rejected.

4.5.2 Chemical and physical tests. Tests to determine conformance to chemical and physical properties of the propellant supplied shall be conducted in accordance with Table I. The finished propellant shall pass all tests listed in Table I. No additional testing shall be required. Requirements contained herein are not subject to corrections for test tolerances. If multiple determinations are made, results falling within any specified repeatability and reproducibility tolerances may be averaged. For rounding of significant figures, ASTM E29 shall apply to all tests required by this specification.

4.5.3 Testing. Unless otherwise specified by the procuring activity, tests as specified in 3.2 shall be conducted in accordance with ASTM or IP standards, using applicable methods as listed in Table I.

4.6 Rejection and retest. When any sample of the propellant tested in accordance with 4.2 fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected. Rejected material shall not be resubmitted without furnishing full particulars concerning previous rejection and measures taken to overcome defects.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The propellant covered by this specification is intended for use in rocket engines and as a hydraulic fluid medium in rocket engine gimbal systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Grade of the propellant required (1.2).

c. Method of shipment, type, and capacity of containers.

d. Quantity by weight.

e. Packaging requirements (5.1).

f. That one copy of the certificate of analysis, signed by the contractor's representative, listing values obtained on all tests should accompany each shipment delivered to the consignee.

6.3 Definitions.

6.3.1 Barrel. A barrel as specified herein will contain 42 U.S. gallons.

6.3.2 Lot. A lot would consist of one of the following:

a. The propellant produced in not more than 24 consecutive hours from a continuous process that is used to fill shipping containers directly from the process output. A continuous process should be the production of product by continuous input of raw materials and output of finished product by one manufacturer in one plant with no change in manufacturing conditions or materials.

b. The propellant from individual runs of a batch process that is used to fill shipping containers directly from the process output. A batch process should be the production of product by single additions of raw materials that are reacted and purified forming the product.

c. The propellant from either or both the continuous and batch processes that is held in a single storage tank and subsequently withdrawn to fill shipping containers. The product should be homogenous at the time of withdrawal and should not be added to while being withdrawn. After each addition to the storage tank, the contents would constitute a separate lot.

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6.4 Subject term (key word) listing.

Antioxidant
Fuel
Metal deactivator
Rocket engine
RP-1
RP-2

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR4
Navy - AS
Air Force - 68
DLA - PS

Review activities:
Army - AT, MI
Air Force -19

Civil agencies:
NASA - NA

Preparing activity:
Air Force - 68
(Project 9130-2019-004)

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